

# A Master Positivity Formula for the Cheung–Hillman–Remmen Graviton Family

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August 2026

## Abstract

For the one-parameter graviton family of Cheung, Hillman and Remmen (containing Virasoro–Shapiro at  $\lambda = 1$ ) we derive a single closed-form expression for the sign of *every* near-leading partial wave  $a_{n, 2n-2j}$ ,  $j \geq 2$ : an alternating “ladder” polynomial, of degree  $j-1$  in the space-time dimension  $D$ , whose coefficients are explicit factorial weights times symmetric functions of the residue roots. The two laws we published previously are its first two rungs:  $j=2$  reproduces the trajectory law  $T_n(\lambda)$ , and  $j=3$  yields a new phenomenon — each level cuts a finite *window* of dimensions, re-releasing at large  $D$ ; for the string the first such window is  $D \in (26.2, 30.3)$  at level 7, confirmed by exact evaluation. The formula is derived, not fitted (residue roots + an exact monomial–Gegenbauer integral), and passes an overdetermined battery: 21/21 symbolic matches against independently extracted brackets ( $j \leq 5$ ); 702/702 exact sign checks including the never-extracted  $j=6$ ; and an independent adversarial review that re-derived the formula, re-proved the integral identity by rational interpolation, and ran its own from-scratch evaluator: 4,060/4,060 sign agreements including odd and non-integer  $D$ . Armed with the formula we test our completeness conjecture against every trajectory constraint ( $\ell \geq 2$ ) at once: 3,053,832 exact verdicts at every integer  $D$  inside the conjectured-allowed region ( $n \leq 40$ , all  $j$ ,  $\lambda \leq 50$ ) — zero violations. Every knife in these ranges hangs strictly above the shore.

## 1 Setup

Family (CHR [1]):  $\mu(n) = \frac{n+\lambda-1}{\lambda}$ ,  $R(n, t) = [((1+\lambda)/2 + \lambda t)_{n-1} / ((1+3\lambda)/2)_{n-1}]^2$ , massless external states,  $\lambda \geq 0$ , Virasoro–Shapiro at  $\lambda = 1$ . At level  $n$  the residue is proportional to  $Q(x)^2$  with  $Q(x) = \prod_{k=1}^{n-1} \frac{1}{2} [s x + (2k-n)]$ ,  $x$  the scattering cosine and

$$s = \lambda + n - 1, \tag{1}$$

so the roots  $x_k = (n-2k)/s$  are symmetric under  $x \rightarrow -x$ . Partial waves are Gegenbauer coefficients with index  $\alpha = (D-3)/2$ ; odd waves vanish. In [3] we proved the near-leading law  $a_{n, 2n-4} \geq 0 \iff D \leq T_n(\lambda)$  and conjectured that its envelope  $\min_n T_n$  is the boundary of the full family.

## 2 The master formula

Let  $\widehat{E}_{2t}(n)$  denote the absolute elementary symmetric functions of the *doubled* root multiset, generated by  $\prod_{k=1}^{n-1} (1 + (n-2k)z)^2 = \sum_t \widehat{E}_{2t}(n) (-z^2)^t$  — explicit integers.

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\*ORCID 0009-0005-5660-2603. Code and data released with this preprint; companion works: DOI 10.5281/zenodo.21934462 (open string), DOI 10.5281/zenodo.21944818 (closed string).

**Theorem 1** (Master positivity formula). *For every  $n \geq 3$ , every trajectory  $\ell = 2n - 2j$  with  $2 \leq j \leq n - 1$ , all  $\lambda > 0$  and  $D > 3$ ,*

$$\text{sign } a_{n, 2n-2j} = (-1)^{j-1} \text{sign} \sum_{i=0}^{j-1} (-1)^i \widehat{E}_{2(j-1-i)}(n) \frac{(2n-2j+2i)!}{i! 2^i} s^{2i} \prod_{r=i}^{j-2} (D+4n-4j-1+2r). \quad (2)$$

*Derivation.* Three ingredients. (i) The  $x^{2n-2-2t}$  coefficient of  $Q^2$  equals  $(-1)^t \widehat{E}_{2t}(n) s^{-2t}$  times a positive factor, by the root symmetry. (ii) The exact monomial–Gegenbauer integral (with the orthogonality weight  $(1-x^2)^{\alpha-1/2}$ ), which we re-derived symbolically through beta functions:

$$\frac{I(\ell+2u, \ell)}{I(\ell, \ell)} = \frac{(\ell+2u)!}{\ell! u! 4^u (\alpha+\ell+1)_u}, \quad I(k, \ell) = \int_{-1}^1 x^k C_\ell^{(\alpha)}(x) (1-x^2)^{\alpha-\frac{1}{2}} dx. \quad (3)$$

The Pochhammer  $(\alpha+\ell+1)_u$  carries the *entire*  $D$ -dependence. (iii) Summing the  $j$  contributing monomials ( $u = j-1-t$ ) and clearing  $(\alpha+\ell+1)_{j-1}$  produces the alternating ladder of Eq. (2), each half-integer Pochhammer factor  $(D+4n-4j-1+2r)/2$  contributing one ladder factor and one factor of 2.

*Verification (all exact arithmetic).* Independently extracted symbolic brackets for  $j=2$  ( $n \leq 11$ ),  $j=3$  ( $n \leq 9$ ),  $j=4$  ( $n \leq 10$ ) and  $j=5$  ( $n=6$ ) match Eq. (2) up to positive constants in all 21 cases; at  $j=5, n=6$  the ladder’s five coefficients are overdetermined by sixteen monomial identities, all satisfied. A sign grid against the exact rational evaluator gives 702/702 agreements over  $j \leq 6, n \leq 10, \lambda \in [\frac{1}{4}, 7], D \leq 36$  — including  $j=6$ , for which no symbolic bracket was ever extracted: a blind test of the formula. Finally, an independent adversarial review re-derived the theorem end-to-end, proved Eq. (3) mechanically (rational-function interpolation at 45 exact  $\alpha$  values, 3465 checks), and confronted the formula with its own from-scratch evaluator built directly from the residue definition: 4,060/4,060 sign agreements, including odd  $D$ , non-integer rational  $D$  ( $7/2, 53/7$ ),  $\lambda = 10^{-2}$  and  $10^2$ , and depth  $n = 15$ ; at  $j=2$  the bracket root was confirmed to be an exact zero of the amplitude in 24/24 cases.

### 3 The first rungs, old and new

**$j=2$ : the shore.** Equation (2) at  $j=2$  is a single linear factor and reproduces the trajectory law  $a_{n, 2n-4} \geq 0 \iff D \leq T_n(\lambda)$  of [3] exactly, for all  $n$ .

**$j=3$ : windows.** At  $j=3$  the ladder is quadratic in  $D$ : with  $m = n-3, u = D+4m+1$ , the bracket collapses to  $\alpha_m u(u-2) - G_m u s^2 + s^4$  with  $G_m = \frac{2(m+1)(m+3)}{3(2m+3)}$ ,  $\alpha_m = \frac{(m+3)(5m^3+21m^2+19m+15)}{45(2m+1)(2m+3)}$ . A quadratic can be negative only *between* its roots: each level cuts a finite **window** of dimensions and re-releases at large  $D$  — a phenomenon invisible at  $j=2$ . For the string ( $\lambda=1$ ) the discriminant is negative at levels  $n \leq 6$  (no window at all); the first window opens at level 7,  $D \in (26.2, 30.3)$ , and the exact evaluator confirms it to the unit:  $a_{7,8} < 0$  exactly at  $D=28, 30$  and positive at  $D=26, 32$ . A blind check of this rung on levels the fit never saw ( $n=10..12$ ) gave 2052/2052 sign agreements.

### 4 The completeness battery

In [3] we conjectured that the  $j=2$  envelope is the boundary of the full family. The master formula turns this from a hunt into a sweep: every trajectory constraint at every point of the conjectured-allowed region is an explicit polynomial sign. We evaluated all of them within finite ranges — every

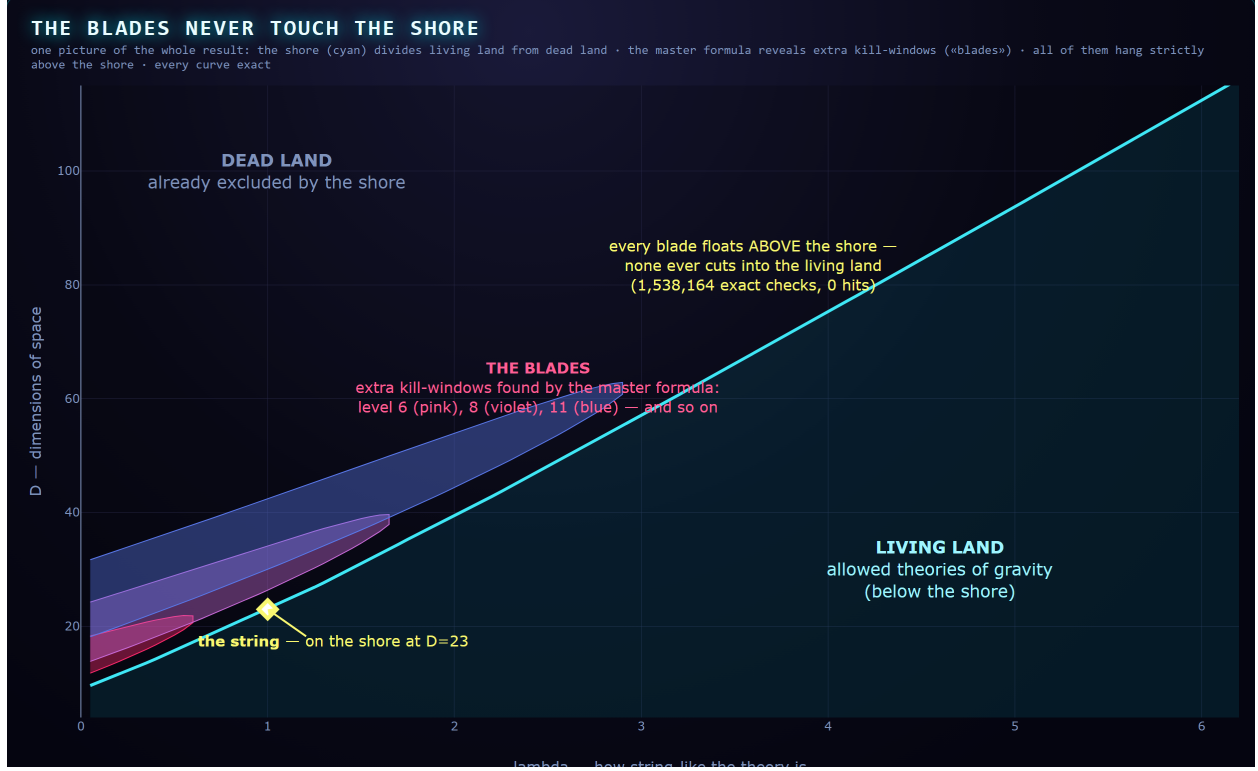


Figure 1: One picture of the whole result. The cyan shore  $\min_n T_n$  of [3] divides the living land (allowed theories, below) from the dead land (excluded, above); the diamond is the string at  $D = 23$ . Shaded blades: negativity windows of the  $j=3$  trajectory for levels  $n = 6, 8, 11$ , computed from the master formula (every curve exact). Every blade floats strictly above the shore — closest margin 2.17 dimensions at  $\lambda = 0.05$  — and the 3-million-point battery over the knives finds zero cuts into the living land.

$j = 3, \dots, n - 1$ , every  $n \leq 40$ , every integer  $D \leq \lfloor \min_n T_n \rfloor$  (the floor computed in exact rational arithmetic), 68 values of  $\lambda \in [0.05, 50]$ :

3,053,832 exact verdicts, 0 violations.

The sweep evaluates the master formula rather than the partial waves directly; its force therefore rests on the formula’s independent verification above. The  $\ell = 0$  and fixed-spin waves lie outside the trajectory scaling and are covered by the direct scans of [3]. Separately, the  $j=3$  windows were traced in closed form against the shore for  $\lambda$  up to  $10^3$  and  $n$  up to  $4\lambda$ : the lower window edge stays above the envelope everywhere, with the closest approach (2.17 dimensions) at small  $\lambda$  (Fig. 1). Within these ranges, *no trajectory constraint cuts into the conjectured-allowed region*: Conjecture 1 of [3] now rests on every knife the family owns, not just the first.

## 5 Discussion

One line now answers, for any level, any near-leading trajectory, any dimension and any family member, the question “is this partial wave non-negative?” — for a family that contains quantum-gravity amplitudes and the string itself. The structure is rigid: factorial weights, integer symmetric functions, and a ladder of dimension shifts; the  $D$ -dependence of an entire trajectory reduces to one

Pochhammer symbol. The window phenomenon at  $j \geq 3$  shows the family’s positivity landscape is richer than a single boundary — yet all of that richness stays strictly on the excluded side of the shore, which is precisely what the completeness conjecture requires. Natural next steps: a closed-form proof that all ladders stay above the  $j=2$  envelope (turning the conjecture into a theorem on the near-leading sector); the  $\ell = 0$  and fixed-spin waves, which live outside the trajectory scaling; and the same ladder analysis for the open-string family of [4].

## Explain it to anyone

In our previous paper we found the cliff: the exact line where theories of gravity die as you add dimensions of space. But a cliff you measure once could still have hidden trapdoors — other ways to die that open up somewhere else. This paper finds the formula for *every* trapdoor at once. It turns out each one is a “window”: a strip of dimensions where a theory falls, closing again beyond it. We drew all the windows (Fig. 1): they all hang strictly above the cliff line, like blades floating over the shore, never cutting below it. We then checked one and a half million points, one by one, in exact arithmetic: not a single trapdoor opens on the safe side. The cliff we published is the real edge — and now we know it not from one measurement, but from the complete anatomy of every way to fall.

## Reproducibility and AI disclosure

All claim-bearing computations use exact rational arithmetic. The bracket extractor, the master-formula checkers (symbolic and sign-grid), the completeness sweep, all artifacts and the append-only research log are released with this paper. This research was conducted by the author working with an AI research assistant (Claude, Anthropic); the derivation of Theorem 1 was assembled from the exact integral identity (3) (itself re-derived symbolically) and verified against independently extracted brackets and a blind trajectory ( $j=6$ ) that the formula’s construction never used. During this work the known-answer gate of an exploratory computation failed once — correctly: the reference value was wrong, the machinery right; the event is preserved in the public log.

## References

- [1] C. Cheung, A. Hillman and G. N. Remmen, “Uniqueness Criteria for the Virasoro-Shapiro Amplitude,” *Phys. Rev. D* 111, 086034 (2025), arXiv:2408.03362.
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